
3.5.1 Stimuli, both internal and external, are detected and lead to a response.

Survival and response	Organisms increase their chance of survival by responding to changes in their environment. Tropisms as responses to directional stimuli that can maintain the roots and shoots of flowering plants in a favourable environment. Taxes and kinesis as simple responses that can maintain a mobile organism in a favourable environment. A simple reflex arc involving three neurones. The importance of simple reflexes in avoiding damage to the body.
Control of heart rate	The role of chemoreceptors and pressure receptors, the autonomic nervous system and effectors in controlling heart rate.
Receptors	The basic structure of a Pacinian corpuscle as an example of a receptor. The creation of a generator potential on stimulation. The Pacinian corpuscle should be used as an example to illustrate the following. • Receptors only respond to specific stimuli • Stimulation of receptor membranes produces deformation of stretch-mediated sodium channels, leading to the establishment of a generator potential. Differences in sensitivity and visual acuity as explained by differences in the distribution of rods and cones and the connections they make in the optic nerve.
